

Objective To determine the value of acceleration due to gravity and radius of gyration using bar pendulum.

Apparatus required Bar pendulum, Stopwatch and meter scale.

Formula used:-

A The general formula of the time period for bar pendulum is given by following equation.

$$T = 2\pi \sqrt{\frac{\frac{k^2 + l^2}{L}}{g}} = 2\pi \sqrt{\frac{L_2 + L_1}{g}}$$

$$g = \frac{4\pi^2(L_1 + L_2)}{T^2} = \frac{4\pi^2 L}{T^2} \quad \text{--- (1)}$$

where,  $L_1$ : distance between C.G. and suspension point,  
 $L_2$ : distance between suspension and oscillation point.

$$L = L_1 + L_2 = L + \frac{k^2}{L}$$

$g$ : acceleration due to gravity,  $T$ : time period.

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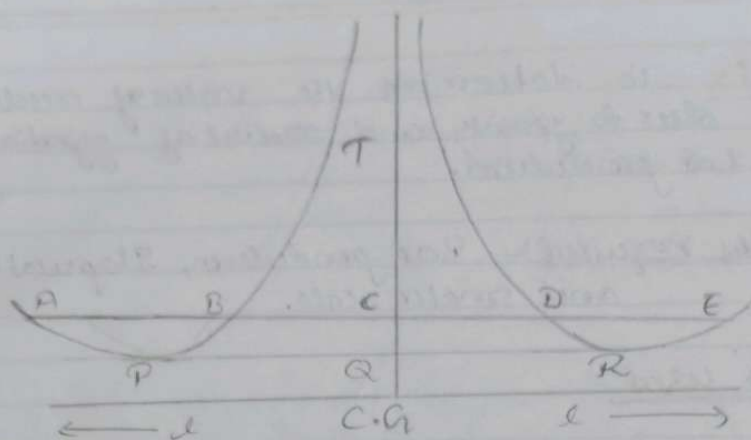


Fig. 2

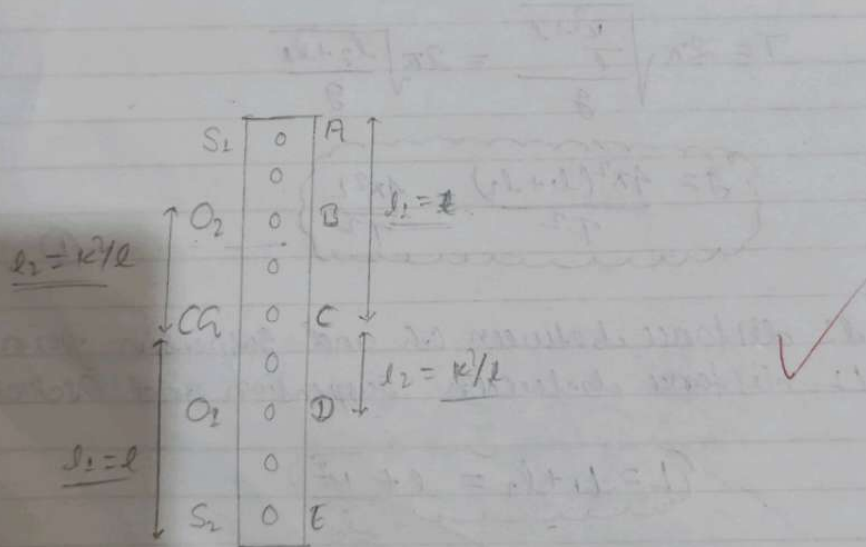


Fig. (1)



B. The time period is minimum when  $l = \pm k$ , in this situation the equation (1) becomes as:

$$T_{\min} = 2\pi \sqrt{\frac{2k}{g}}$$

$$g = \frac{8\pi^2 k}{T_{\min}^2}$$

(2)

or

where,  $k$ : radius of gyration,  $T_{\min}$ : minimum time period.

The value of 'g' can be calculated using equations (1) & (2).

The values of  $L$ ,  $T$ ,  $k$  and  $T_{\min}$  are obtained using graph between  $T$  and  $L$  for bar pendulum which is shown in following figure.

From Figure (1) and (2)

(a)  $L_1 = AC + CD$ ,  $L_2 = EC + EB$  and  $L = (L_1 + L_2)/2$ ,  $T$  = time at

(b)  $k = (PC + QR)/2$  and  $T_{\min}$  = time at Q

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Calculation from graph,

$$L = (AD + EB) / 2 = \frac{62 + 63}{2} = 62.5 \text{ cm} \quad T = \dots \text{ sec},$$

$$K = PR / 2 = \frac{60}{2} = 30 \text{ cm} \quad T_{\min} = \dots \text{ sec},$$

$$L_1 = (AC + CE) / 2 = \dots 45$$

$$L_2 = (BC + CD) / 2 = \dots 17.5$$

$$1. \quad g_1 = \frac{4\pi^2 L}{T^2}$$

$$2. \quad g_2 = \frac{8\pi^2 K}{T_{\min}^2}$$

$$3. \quad g = \frac{g_1 + g_2}{2}$$

$$(1) \quad g_1 = \frac{4\pi^2 L}{T^2} = \frac{4 \times 3.14 \times 62.5 \times 10^{-2}}{(1.00)^2} = 8.94 \text{ m/sec}^2$$

$$(2) \quad g_2 = \frac{8\pi^2 K}{T_{\min}^2} = \frac{8 \times 3.14 \times 30 \times 10^{-2}}{(1.55)^2} = 9.9 \text{ m/sec}^2$$

$$(3) \quad g = \frac{g_1 + g_2}{2} = \frac{8.94 + 9.9}{2} = \frac{18.84}{2} = 9.42 \text{ m/sec}^2$$

$$(4) \quad K = \sqrt{L_1 L_2} = \sqrt{45 \times 17.5} = \sqrt{787.5} = 28.000 \approx 28 \text{ cm}$$

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C The radius of gyration can be obtained with following formula.

$$K = \sqrt{I_1 I_2}$$

where,  $I_1 = (AC + CE)/2$ ,  $I_2 = (BC + CD)/2$

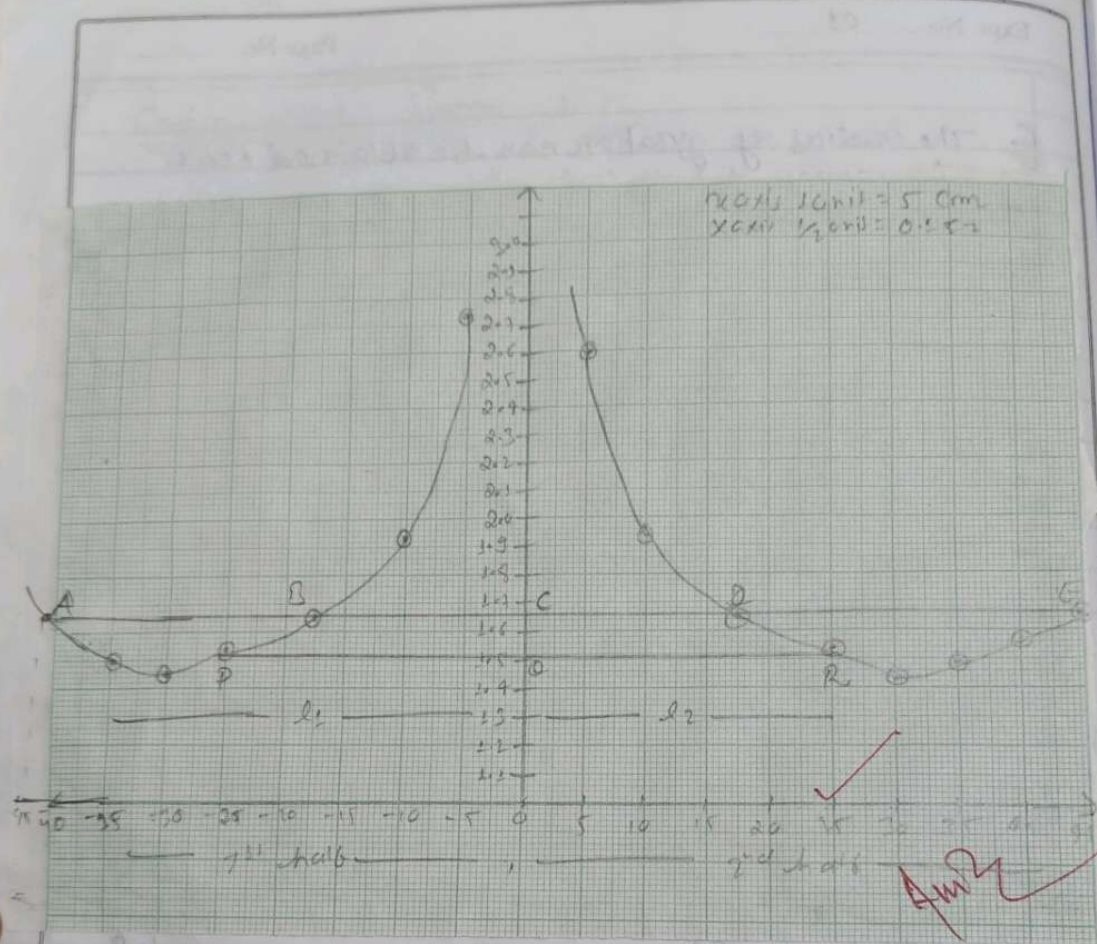
### Observations:-

1. Least count of the stop watch = \_\_\_\_\_ sec
2. Least count of the meter scale = \_\_\_\_\_ cm

### 3. Table for d and T

| S.No.                           | d (cm) | t (time taken for 20 oscillations) | $T = t/20$ |
|---------------------------------|--------|------------------------------------|------------|
| For First half side of the bar  |        |                                    | 6.         |
| 1.                              | 45     | 38.5                               | 1.925      |
| 2.                              | 40     | 38.5                               | 1.925      |
| 3.                              | 35     | 31.5                               | 1.575      |
| 4.                              | 30     | 31                                 | 1.55       |
| 5.                              | 25     | 31                                 | 1.55       |
| 6.                              | 20     | 32                                 | 1.6        |
| 7.                              | 15     | 34                                 | 1.7        |
| 8.                              | 10     | 39                                 | 1.95       |
| 9.                              | 5      | 53                                 | 2.625      |
| For Second half side of the bar |        |                                    |            |
| 10.                             | -5     | 55                                 | 2.75       |
| 11.                             | -10    | 39.5                               | 1.975      |

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|     |     |     |     |
|-----|-----|-----|-----|
| 1.0 | 1.0 | 1.0 | 1.0 |
| 1.1 | 1.1 | 1.1 | 1.1 |
| 1.2 | 1.2 | 1.2 | 1.2 |
| 1.3 | 1.3 | 1.3 | 1.3 |
| 1.4 | 1.4 | 1.4 | 1.4 |
| 1.5 | 1.5 | 1.5 | 1.5 |
| 1.6 | 1.6 | 1.6 | 1.6 |
| 1.7 | 1.7 | 1.7 | 1.7 |
| 1.8 | 1.8 | 1.8 | 1.8 |
| 1.9 | 1.9 | 1.9 | 1.9 |
| 2.0 | 2.0 | 2.0 | 2.0 |
| 2.1 | 2.1 | 2.1 | 2.1 |
| 2.2 | 2.2 | 2.2 | 2.2 |
| 2.3 | 2.3 | 2.3 | 2.3 |
| 2.4 | 2.4 | 2.4 | 2.4 |
| 2.5 | 2.5 | 2.5 | 2.5 |
| 2.6 | 2.6 | 2.6 | 2.6 |
| 2.7 | 2.7 | 2.7 | 2.7 |
| 2.8 | 2.8 | 2.8 | 2.8 |
| 2.9 | 2.9 | 2.9 | 2.9 |
| 3.0 | 3.0 | 3.0 | 3.0 |



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|     |     |      |       |
|-----|-----|------|-------|
| 12. | -15 | 34.4 | 1.72  |
| 13. | -20 | 33   | 1.635 |
| 14. | -25 | 31.5 | 1.56  |
| 15. | -30 | 30.5 | 1.55  |
| 16. | -35 | 31.5 | 1.575 |
| 17. | -40 | 32.5 | 1.61  |
| 18. | -45 | 33   | 1.65  |

Results:The acceleration due to gravity ( $g$ ) =  $\approx 9.7 \text{ m/sec}^2$ Radius of gyration ( $k$ ) = 30 cm (from calculation)Precautions:-

1. The motion of the pendulum should be in a vertical plane. While taking the time, start taking observations after two oscillations to avoid any irregularity of motion.

2. The amplitude of oscillation should be small.

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### Even Semester 2025-26 Examination Admit Card



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| Subject Code | Subject Name                           | Exam Date* | Timings        | AnsBookNo. |
|--------------|----------------------------------------|------------|----------------|------------|
| BAS203       | Engineering Mathematics-II             | 08/05/2026 | 9.30AM-12.30PM |            |
| BAS201       | Engineering Physics                    | 12/05/2026 | 9.30AM-12.30PM |            |
| BEE201       | Fundamentals of Electrical Engineering | 14/05/2026 | 9.30AM-12.30PM |            |
| BCS201       | Programming for Problem Solving        | 16/05/2026 | 9.30AM-12.30PM |            |
| BAS204       | Environment and Ecology                | 19/05/2026 | 9.30AM-12.30PM |            |
| BAS251       | Engineering Physics Lab                |            |                |            |
| BVA251       | Sports and Yoga                        |            |                |            |
| BCE251       | Engineering Graphics & Design Lab      |            |                |            |
| BEE251       | Basic Electrical Engineering Lab       |            |                |            |
| BCS251       | Programming for Problem Solving Lab    |            |                |            |

#### Important Note (Read Carefully)

- 1: Reporting time at exam centre **atleast an hour** before the commencement of exam. **No entry** is allowed after the start of the exam.
- 2: The student should bring the Admit Card along with a Institute Photo Id /Other valid Id for appearing in the examination failing which he/she shall not be allowed to appear in the examination.
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